



Mathematics Part 1 — Cbse Chapter Test — 20 Marks

ANSWER KEY — FOR GRADING

Subject: Mathematics Part 1 | Class: Class 8

Total Marks: 20 | Time Allowed: 0h 40m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	Why was the Mesopotamian system not considered a fully developed place value system? (A) It had no concept of zero (B) Blank spaces created ambiguity in reading numbers (C) It used too many symbols (D) It could not represent large numbers Answer: Correct Option: (B) ► Why? B. The Mesopotamian system had defects that led to confusion—blank spaces were used to indicate missing place values, but the spacing between symbols was not consistent, creating ambiguities. [Ch 3: A Story of Numbers]	1
2	What is the term used for multiplication by parts, as referred to by Brahmagupta? (A) Lilavati (B) Khaṇḍa-guṇanam (C) Brahmasphuṭasiddhānta (D) Ista-gunana Answer: Correct Option: (B) ► Why? B. Khaṇḍa-guṇanam. Brahmagupta referred to the use of the distributive property for multiplication as khaṇḍa-guṇanam, meaning multiplication by parts. [Ch 6: We Distribute, Yet Things Multiply]	1



Q.NO.	QUESTIONS	MARKS
3	Assertion (A): In a mixture of 40 kg containing sand and cement in the ratio 3:1, the weight of sand is 30 kg. Reason (R): The weight of sand is calculated as $(3/(3+1)) \times 40 = 30$ kg. (A) A is false but R is true. (B) A is true but R is false. (C) Both A and R are true, and R is the correct explanation of A. (D) Both A and R are true, but R is NOT the correct explanation of A. Answer: Correct Option: (C) ► Why? C. Both A and R are true, and R is the correct explanation of A. The source explicitly states that the weight of sand is $3/(3+1) \times 40 = 30$ kg. The reason provides the correct formula and calculation. [Ch 7: Proportional Reasoning-1]	1
4	All odd numbers can be written as which of the following? (A) Sum of two consecutive numbers (B) Sum of three consecutive numbers (C) Product of two consecutive numbers (D) Sum of four consecutive numbers Answer: Correct Option: (A) ► Why? A. Sum of two consecutive numbers. All odd numbers can be written as a sum of two consecutive numbers, for example, $7 = 3 + 4$ and $15 = 7 + 8$. [Ch 5: Number Play]	1
5	When two even numbers that are NOT multiples of 4 are added, the result is always: (A) A multiple of 4 (B) An odd number (C) A multiple of 2 but not of 4 (D) Not divisible by 2 Answer: Correct Option: (A) ► Why? A. A multiple of 4. Two even numbers that are not multiples of 4 can be written as $(4p + 2)$ and $(4q + 2)$. Their sum is $4p + 4q + 4 = 4(p + q + 1)$, which is always a multiple of 4. [Ch 5: Number Play]	1
6	Assertion (A): The opposite sides of a rectangle are parallel to each other. Reason (R): The diagonals of a rectangle are perpendicular to each other. (A) A is false but R is true. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) Both A and R are true, and R is the correct explanation of A. (D) A is true but R is false. Answer: Correct Option: (D)	1



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	► Why? D. The assertion is true as the source states that opposite sides of a rectangle are parallel to each other. The reason is false because the source only states that diagonals of a rectangle are equal and bisect each other; perpendicular diagonals are a property of squares and rhombuses, not rectangles. [Ch 4: Quadrilaterals]	
7	What is a trapezium? (A) A quadrilateral with all sides equal (B) A quadrilateral with at least one pair of parallel opposite sides (C) A quadrilateral with opposite sides parallel and equal (D) A quadrilateral with diagonals perpendicular to each other Answer: Correct Option: (B) ► Why? B. A trapezium is defined as a quadrilateral having at least one pair of parallel opposite sides. [Ch 4: Quadrilaterals]	1
8	Given $125^2 = 15625$, what is the value of 126^2? (A) $15625 + 126$ (B) $15625 + 26^2$ (C) $15625 + 253$ (D) $15625 + 251$ (E) $15625 + 51^2$ Answer: Correct Option: (D) ► Why? The chapter explains that consecutive squares differ by consecutive odd numbers. The pattern shows that $(n+1)^2 = n^2 + (2n+1)$. For $n = 125$: $126^2 = 125^2 + (2 \times 125 + 1) = 15625 + 251 = 15876$. Alternatively, using the odd number pattern, the 126th odd number is $2 \times 126 - 1 = 251$, so $126^2 = 125^2 + 251$. ► Step-by-step: - Step 1: Use the formula $(n+1)^2 = n^2 + 2n + 1$ - Step 2: For $n = 125$: $126^2 = 125^2 + 2 \times 125 + 1$ - Step 3: $126^2 = 15625 + 250 + 1 = 15625 + 251$ - Step 4: Therefore, option (iv) $15625 + 251$ is correct [Ch 1: A Square and A Cube]	1
Section B Attempt all 3 questions. Each carries 2 mark(s).		
9	Write the formula for the nth odd number and use it to find the 36th odd number. Answer: The nth odd number is $2n - 1$. Therefore, the 36th odd number is $2(36) - 1 = 72 - 1 = 71$. ► Why? This question tests recall of the formula for the nth odd number and its direct application, both explicitly stated in the source material.	2



Q.NO.	QUESTIONS	MARKS
	[Ch 1: A Square and A Cube]	
10	List any two properties of a rectangle. Answer: A rectangle has four properties: (1) All angles are 90°. (2) Opposite sides are equal in length. (3) Opposite sides are parallel to each other. (4) Diagonals are equal in length and bisect each other. Any two of these can be stated as properties. ► Why? This question tests recall of the properties of rectangles listed in the source material. [Ch 4: Quadrilaterals]	2
11	For 120 students, a cook makes 15 kg of rice. If only 80 students attend, how much rice should be made using proportional reasoning? Answer: The ratio of students to rice must remain proportional. The factor of change is $80/120 = 2/3$. Multiplying the rice quantity by this factor: $15 \times (2/3) = 10$ kg. The cook should make 10 kg of rice to avoid waste. ► Why? This tests the factor of change method for solving proportional reasoning problems, as demonstrated in the mid-day meal example. [Ch 7: Proportional Reasoning-1]	2
Section C Attempt 2 out of 3.		
12	Define a square. Explain why every square is also a rectangle. Answer: Definition: A square is a quadrilateral in which all angles are equal to 90°, and all sides are of equal length. Explanation — Why Every Square is a Rectangle: Step 1: A rectangle is defined as a quadrilateral with all angles equal to 90°. Step 2: A square has all angles equal to 90°. Step 3: Since a square satisfies the definition of a rectangle, every square is a rectangle. Additional Property: A square has an additional property that all its sides are equal, which is not required for a general rectangle. Conclusion: Thus, every square is a rectangle, but not every rectangle is a square. Example: A chess board consists of squares, each of which is both a square and a rectangle.	3



Q.NO.	QUESTIONS	MARKS
	► Why? The answer defines a square correctly and explains the subset relationship between squares and rectangles as stated in the source. ► Step-by-step: 1. Step 1: Define a square as a quadrilateral with all angles 90° and all sides equal. 2. Step 2: Recall that a rectangle requires all angles to be 90°. 3. Step 3: Conclude that since a square has all angles 90°, it satisfies the rectangle definition. [Ch 4: Quadrilaterals]	
	OR What pattern do you observe when adding consecutive odd numbers starting from 1? Show how this pattern gives the first five square numbers. Answer: Pattern: The sum of consecutive odd numbers starting from 1 equals a perfect square. Demonstration: $1 = 1 = 1^2$ $1 + 3 = 4 = 2^2$ $1 + 3 + 5 = 9 = 3^2$ $1 + 3 + 5 + 7 = 16 = 4^2$ $1 + 3 + 5 + 7 + 9 = 25 = 5^2$ From this, we observe that the sum of the first n odd numbers is n^2. Alternatively, every square is a sum of successive odd numbers starting from 1. This relationship provides a visual and numerical way to understand perfect squares. ► Why? The answer establishes the fundamental relationship between consecutive odd numbers and perfect squares, which is explicitly derived from the patterns shown in the source material. ► Step-by-step: 1. Step 1: Write the first five square numbers: 1, 4, 9, 16, 25 2. Step 2: Express each as a sum of consecutive odd numbers starting from 1 3. Step 3: Observe the pattern: $1 = 1^2$, $1+3 = 2^2$, $1+3+5 = 3^2$, and so on 4. Step 4: Conclude that the sum of first n odd numbers equals n^2 [Ch 1: A Square and A Cube]	
13	State the three rules for parity when adding or subtracting odd and even numbers. Using these rules, determine whether $43 + 37$ is even or odd without calculating the sum.	3



Q.NO.	QUESTIONS	MARKS
	Answer: The Three Parity Rules: 1. $\text{odd} \pm \text{odd} = \text{even}$ 2. $\text{even} \pm \text{even} = \text{even}$ 3. $\text{odd} \pm \text{even} = \text{odd}$ To determine whether $43 + 37$ is even without calculating: We first identify that 43 is an odd number and 37 is also an odd number. According to the first rule, when we add two odd numbers, the result is always even. Answer: $\therefore 43 + 37$ is an even number. Similarly, for $672 - 348$: The expression $672 - 348$ involves even – even, which also gives an even result. ► Why? The answer correctly states all three parity rules from the source and applies the appropriate rule ($\text{odd} + \text{odd} = \text{even}$) to determine the parity of $43 + 37$. ► Step-by-step: 1. Step 1: Recall the three parity rules: $\text{odd} \pm \text{odd} = \text{even}$, $\text{even} \pm \text{even} = \text{even}$, $\text{odd} \pm \text{even} = \text{odd}$. 2. Step 2: Identify that 43 and 37 are both odd numbers. 3. Step 3: Apply the rule '$\text{odd} \pm \text{odd} = \text{even}$' to conclude that $43 + 37$ is even. [Ch 5: Number Play]	

